



77 Massachusetts Ave, 8-313, Cambridge, MA 02139, United States

☎ (+1) 617-803-0776 | ✉ ljing@mit.edu | 🌐 jingli.io | 📱 [jingli9111](https://www.linkedin.com/in/jingli9111)

Education

Massachusetts Institute of Technology

Cambridge, MA

PH.D. CANDIDATE IN PHYSICS

2014 - Present

- Research in Deep Learning and Applied Physics.
- Coursework in CS and ML: Design and Analysis of Algorithms, Artificial Intelligence, Machine Learning, Statistical Learning Theory, Computer Vision, Natural Language Processing

Peking University

Beijing, China

B.S. IN PHYSICS, B.A. IN ECONOMICS(DUAL MAJOR)

2010 - 2014

Research in Quantum Information and Quantum Computation

Selected Honor and Award

2010 **Gold Medal**, 41th International Physics Olympiad (IPhO)

Zagreb, Croatia

Experience

Deep Learning and Applied Physics Group in MIT

Cambridge, MA

RESEARCH ASSISTANCE, ADVISOR: MARIN SOLJAČIĆ

2016 - Present

- Applied a variety of machine learning methods to physics, e.g. photonics inverse design.
- Optimized architecture of optical neural network chip; developed hardware level training algorithms.
- Developed efficient unitary recurrent neural network.

Center for Ultracold Atoms in MIT

Cambridge, MA

RESEARCH ASSISTANCE, ADVISOR: WOLFGANG KETTERLE

2014 - 2016

- Built high power laser tapered amplifier; designed analog PID circuits used to generally stabilize laser power.
- Developed atomic physics image software system for AMO physics.

Quantum Information Group in Chinese Academy of Sciences

Beijing, China

UNDERGRADUATE RESEARCHER, ADVISOR: HENG FAN

2011 - 2014

- Investigated optimization problems in quantum cloning and quantum cryptography.
- Developed methods to design entanglement states to reduce measurement error in quantum metrology.

Qidian Technology

Beijing, China

QUANTITATIVE RESEARCHER AND TRADER

2012 - 2014

- Collected massive data from public social media website including LinkedIn, Twitter, Facebook and Weibo.
- Built natural language models to evaluate employee performances in NASDAQ high-tech companies.
- Established financial models using deep neural network to predict company performances given employee performances.
- Achieved high profit rate(20% per year)

Selected Publications

2017	Gated Orthogonal Recurrent Units: On Learning to Forget , Li Jing, Caglar Gulcehre, John Peurifoy, Yichen Shen, Max Tegmark, Marin Soljačić, Yoshua Bengio	AAAI under review
2017	Tunable Efficient Unitary Neural Networks (EUNN) and their application to RNNs , Li Jing, Yichen Shen, Tena Dubček, John Peurifoy, Scott Skirlo, Yann LeCun, Max Tegmark, Marin Soljačić	ICML
2014	Quantum Cloning Machines and the Applications , Heng Fan, Yi-Nan Wang, Li Jing, Jie-Dong Yue, Han-Duo Shi, Yong-Liang Zhang, Liang-Zhu Mu	Phys. Rep.

Skills

Deep Learning Frameworks	Tensorflow, PyTorch, Theano
Language	English, Mandarin
Programming	Python, C++, Matlab, Lua
Full Stack Developer	Javascript, Django, AngularJS, NodeJS, Less, PHP, Swift.